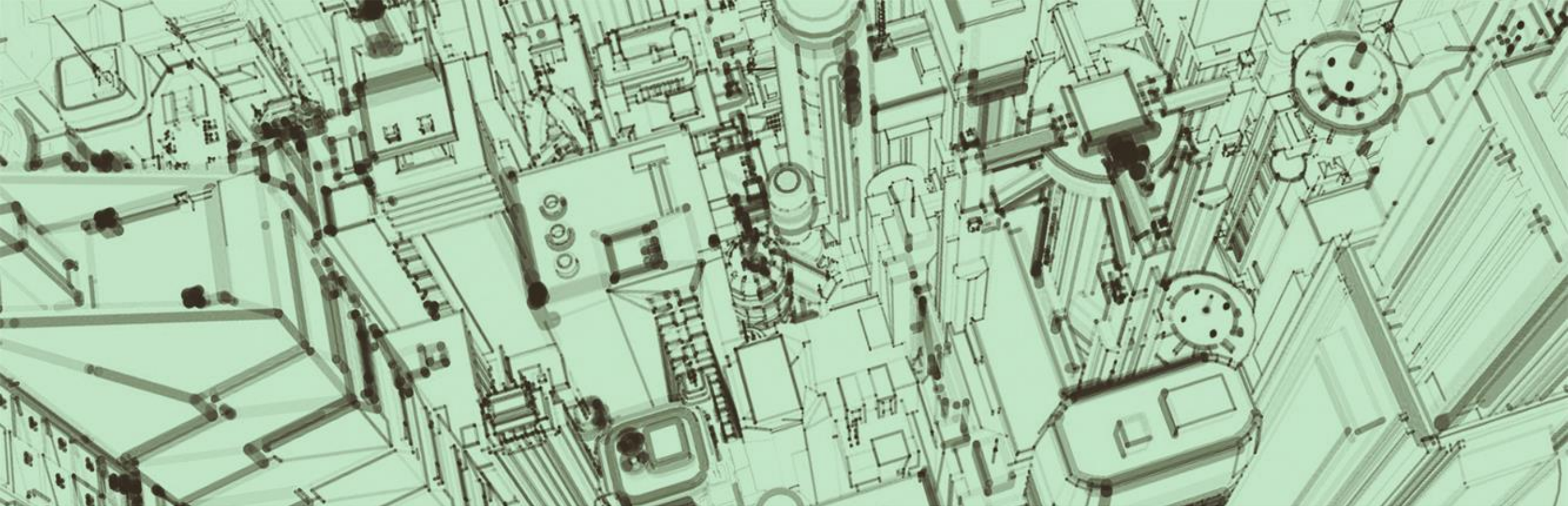


Smart City

Designing a Level 6 IoT System

What is a Smart City?

Using digital technology to collect data and operate services in an urban area.

Integration of people, technology, and processes to connect and interact across different sectors.

ICT

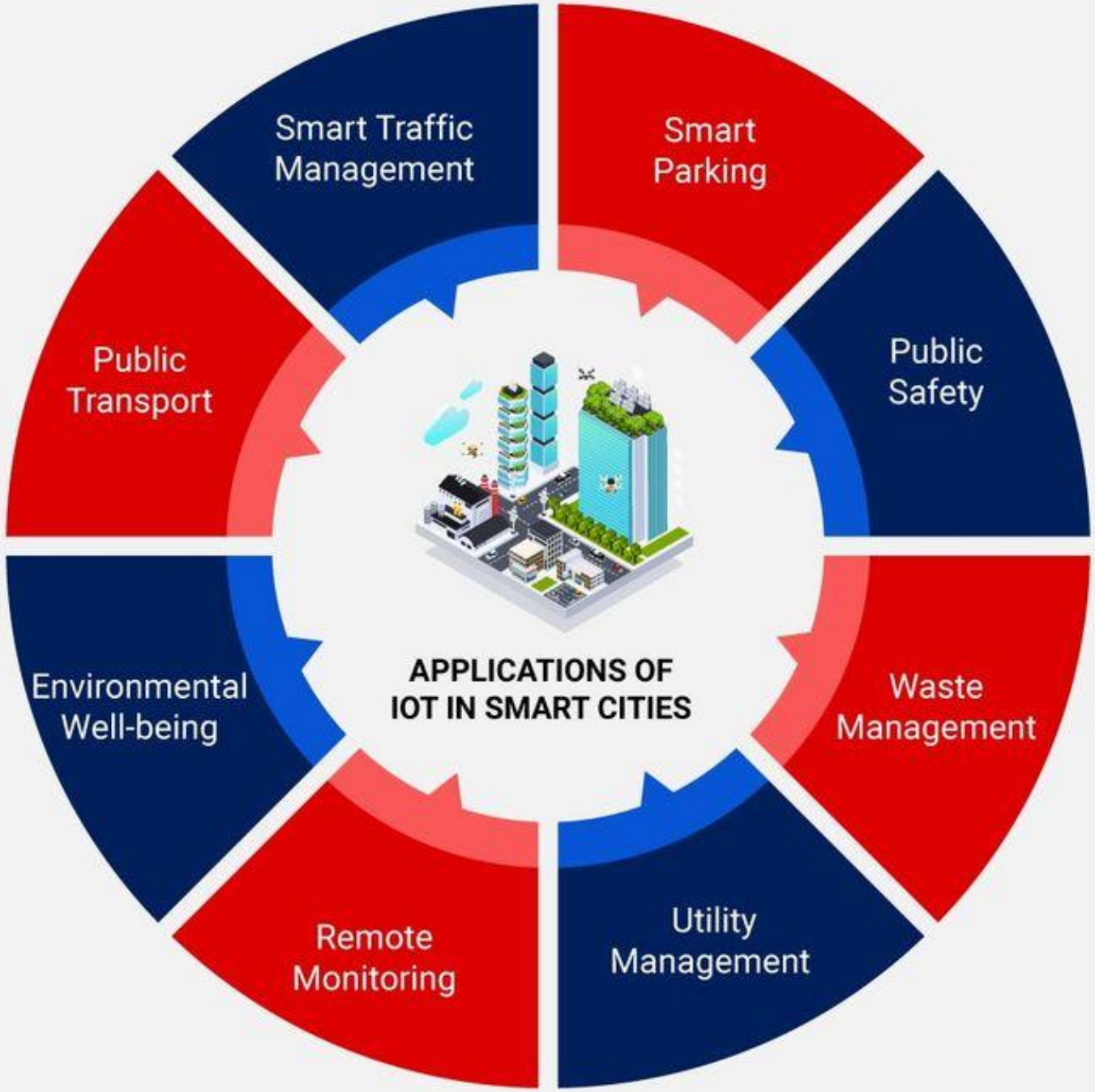
IoT

Automation

AI

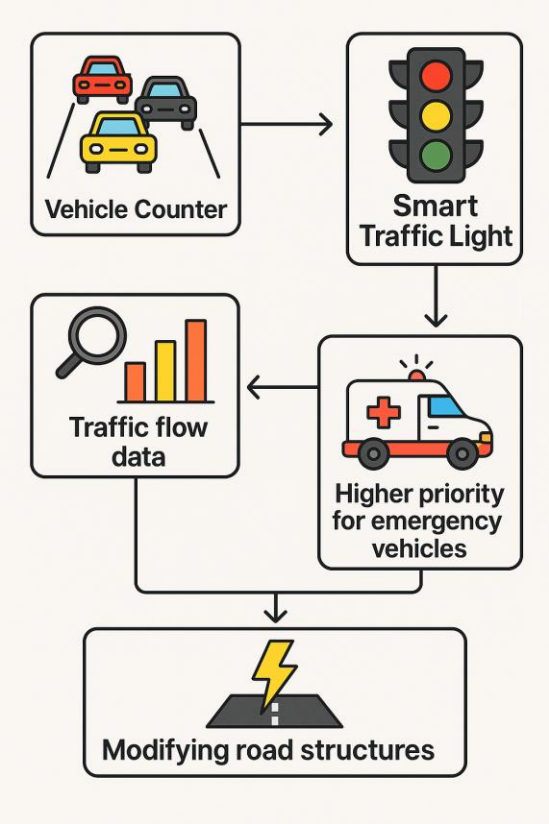
Components of a Smart City

Traffic and Accidents	Environment
Parking	Crime and Threat Management
Advertising	Infrastructure
Energy	Communication



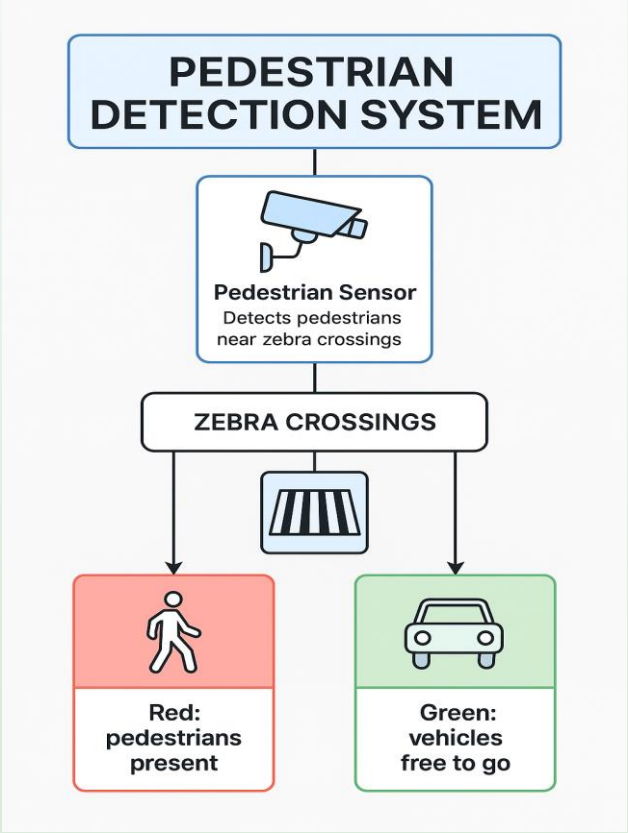
Traffic Management

Intelligent Traffic Lights



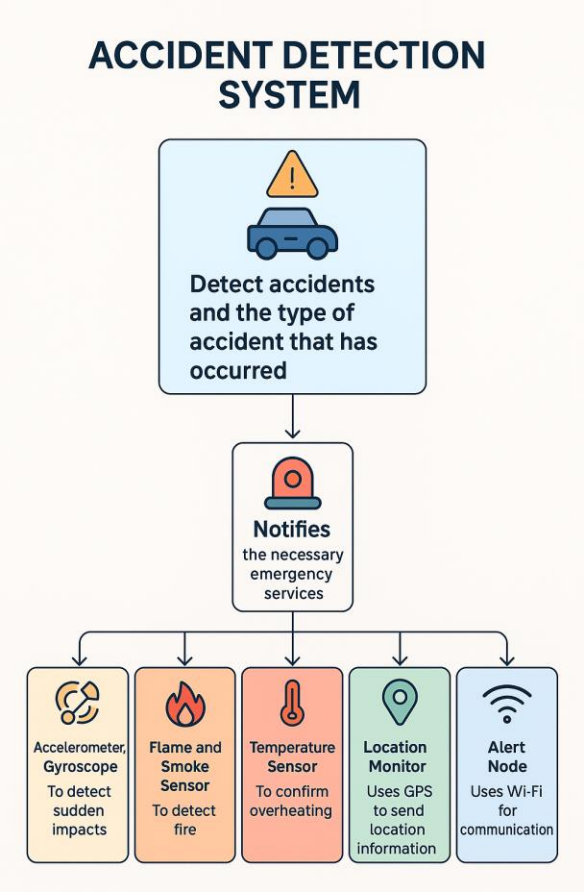
Device	Purpose
Vehicle Counter	To measure vehicle density in each lane
Smart Timer	To set signal timer based on traffic
RFID Reader	To detect emergency vehicles
Smart Controller	To change traffic lights dynamically
Alert Node	Periodically sends traffic summary reports

Smart Zebra Crossings



Device	Purpose
Smart Camera	To detect pedestrians
Motion Sensor	To detect motion on the crossing
Ultrasonic Sensor	To detect incoming vehicles
Controllers	Alters colour of the zebra crossing

Accident Detection and Response

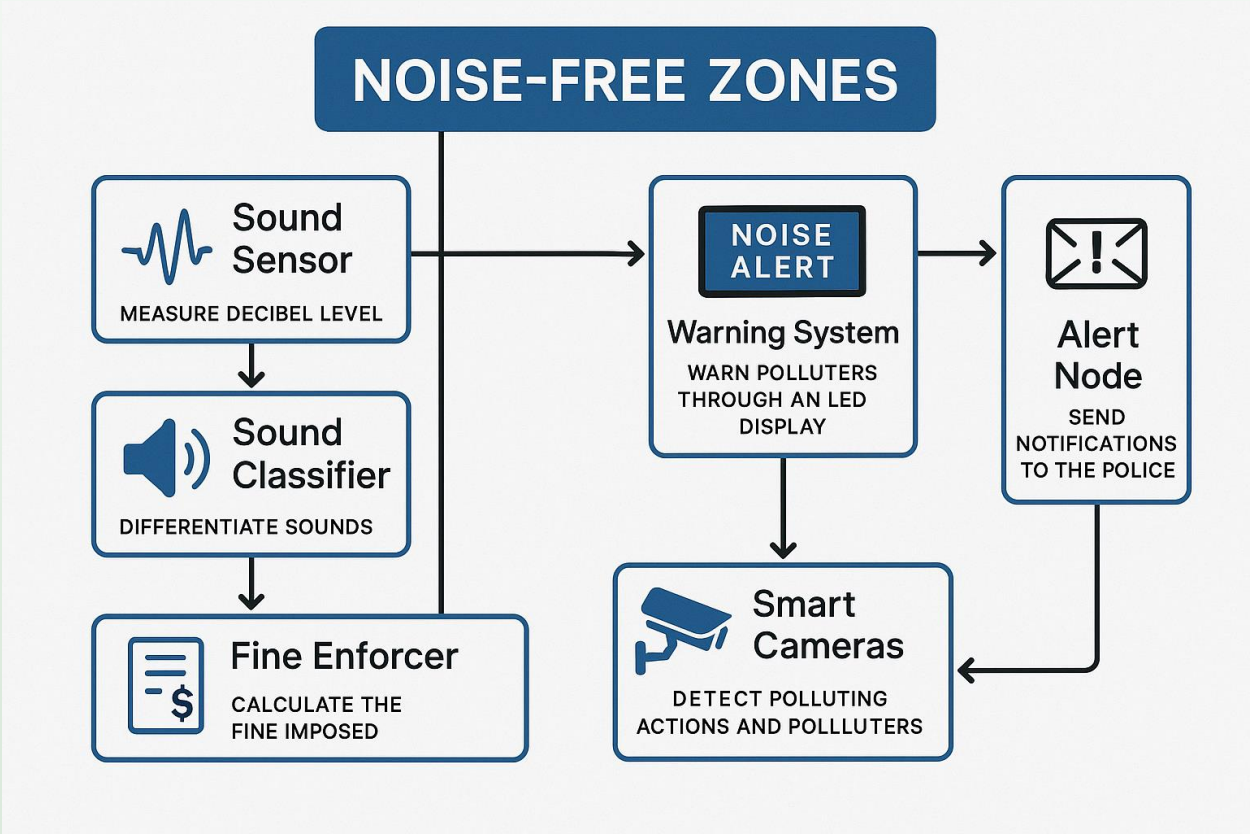


Device	Purpose
Accelerometer, Gyroscope	To detect sudden impacts
Flame and Smoke Sensor	To detect fire
Temperature Sensor	To confirm overheating
Location Monitor	Uses GPS to send location information
Microphone	To detect sudden, high-decibel noises
Alert Node	Uses Wi-Fi for communication

Public Well-being

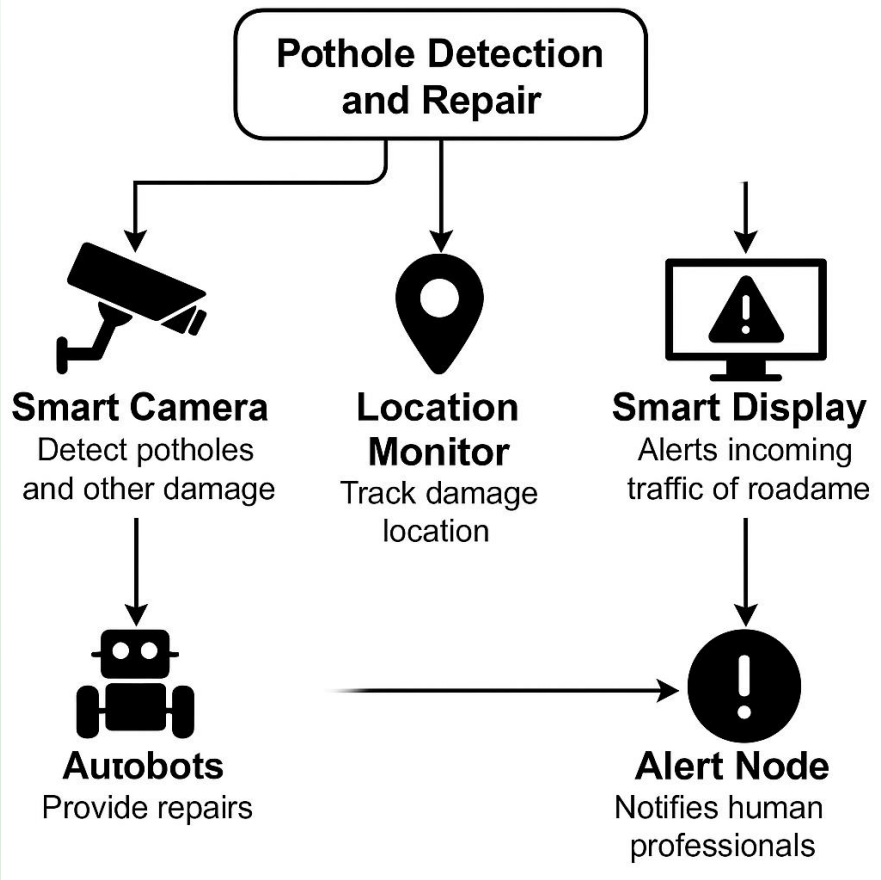
Noise-Free Zones

- A system that monitors noise pollution in areas such as schools and hospitals.
- Display alerts if the noise level exceeds a prescribed limit.
- Store information about vehicle or whatever it is that caused the pollution.
- Check for repetitive actions and fine accordingly.



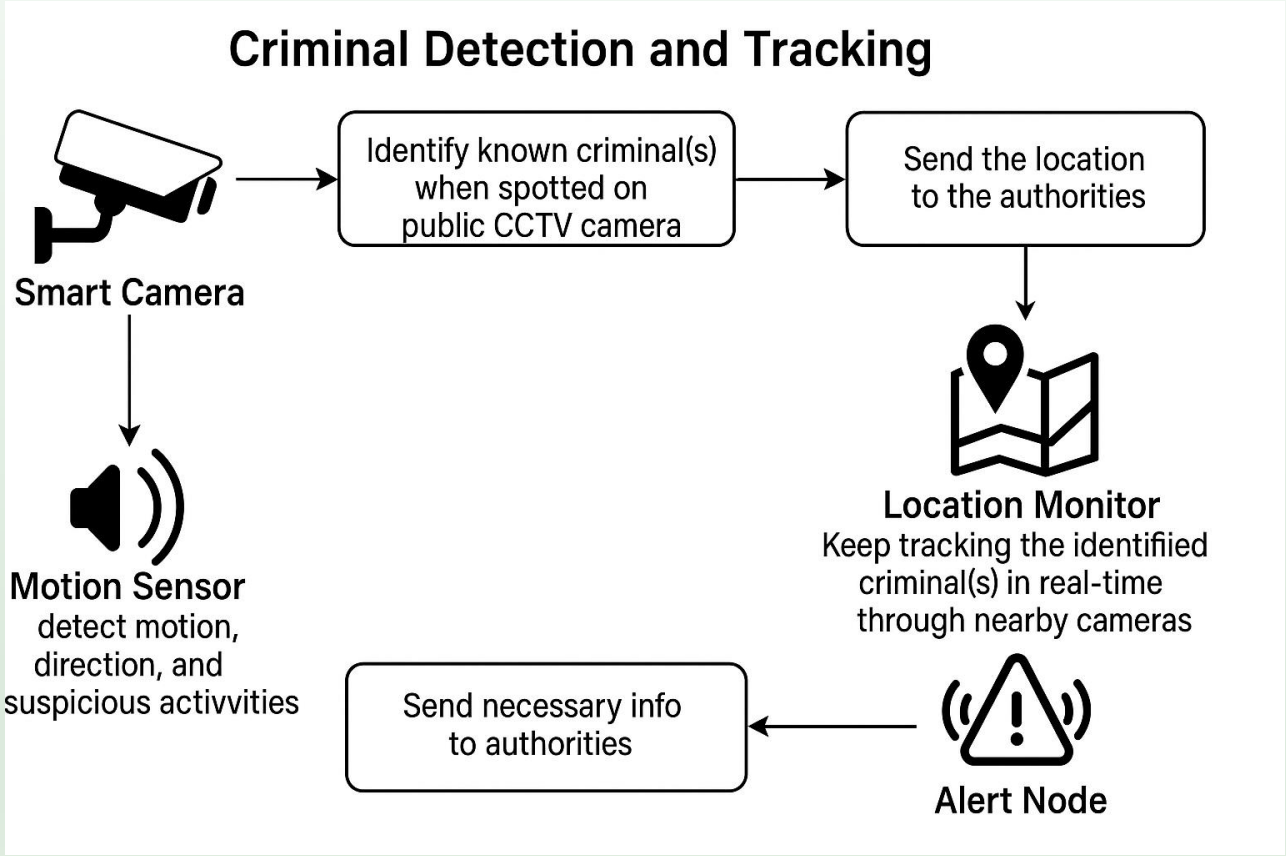
Pothole Detection and Repair

- Detects potholes and similar damages in roads.
- Initiate repair mechanisms using automated bots.
- Perform repairs only when there is no roadside disturbance.
- Keep track of location information for analytics.



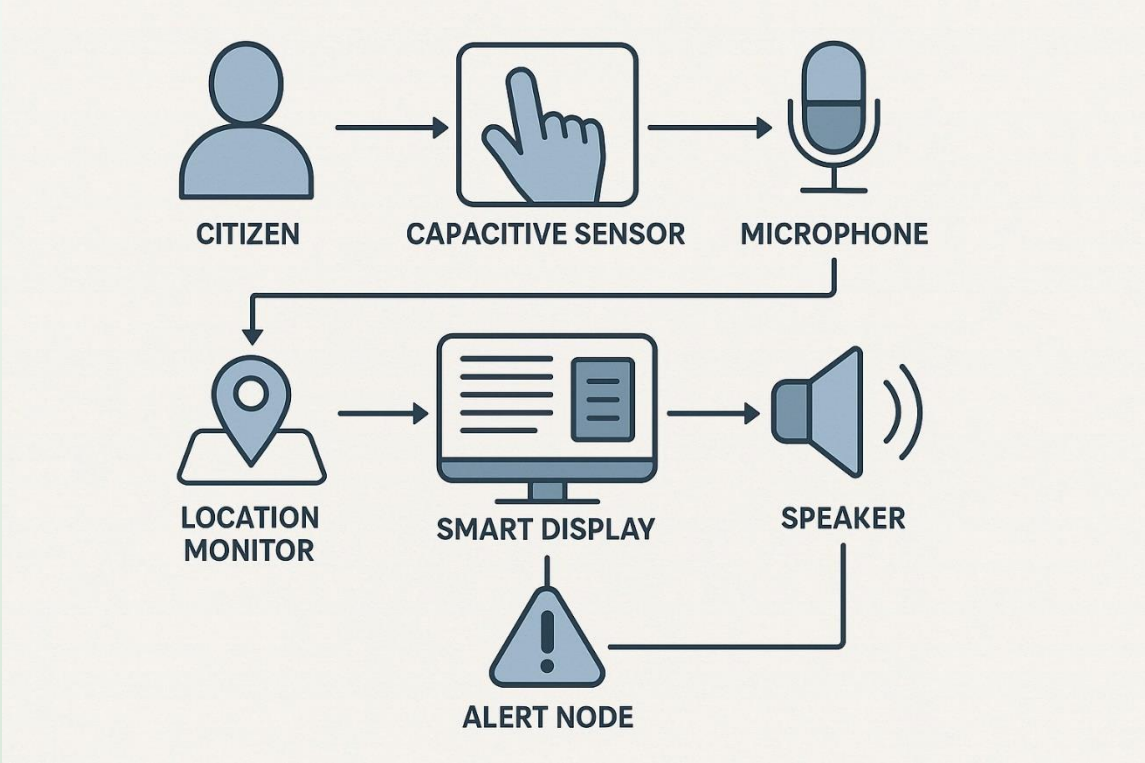
Criminal Detection and Tracking

- Identify known criminal(s) when spotted on a public CCTV camera.
- Send the location to the authorities.
- Keep tracking the identified criminal(s) in real-time through nearby cameras.



Digital Citizen

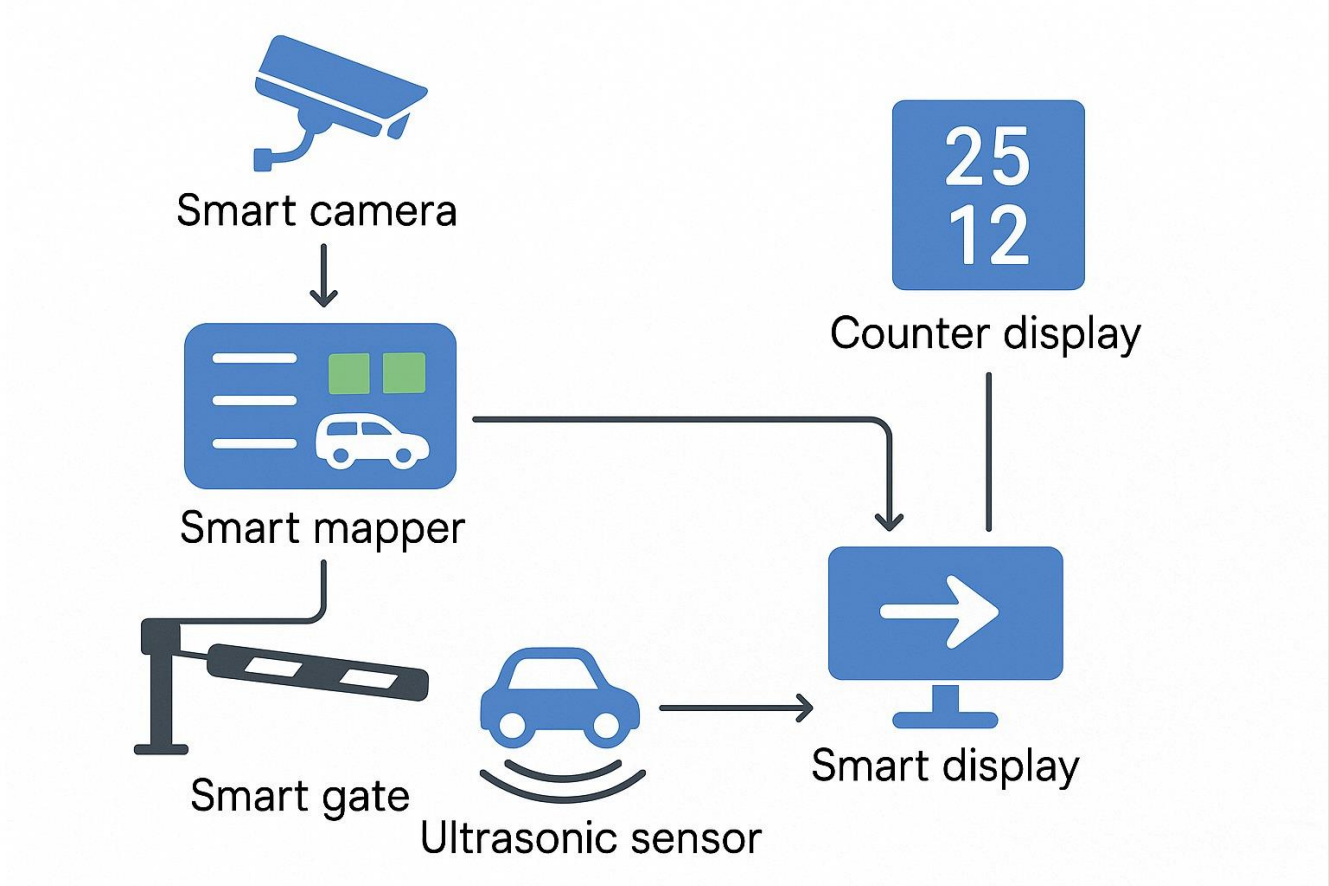
- Enables citizens to post complaints on a digital platform.
- The truth of the complaint is verified and the necessary officials are informed.
- Trustworthy user, multiple corroborations are indicators of truth.
- Manual action is then taken.



Transport

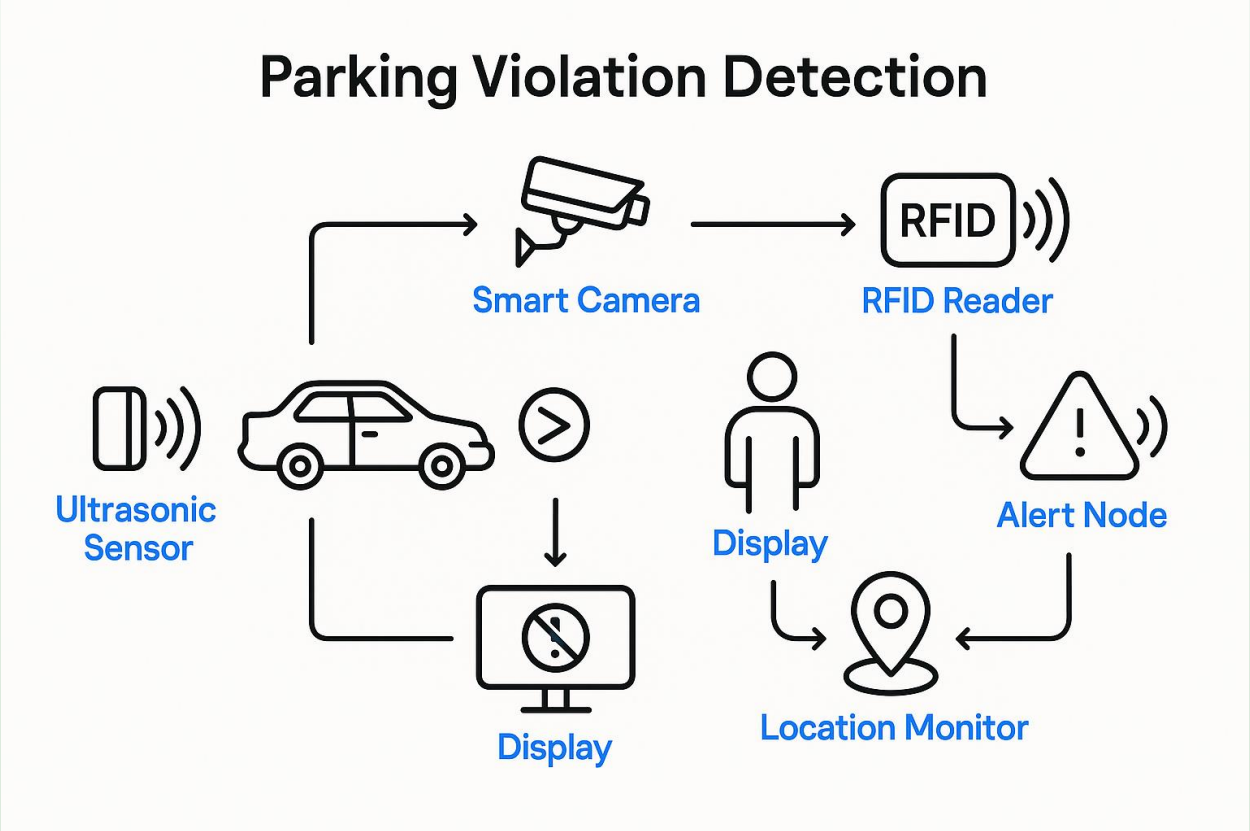
Parking Guidance

- Parking lots that indicate where a vehicle should park through smart displays .
- Analyses already filled spots, detects the type of vehicle, and accordingly provides directions.
- Communication between multiple smart displays.

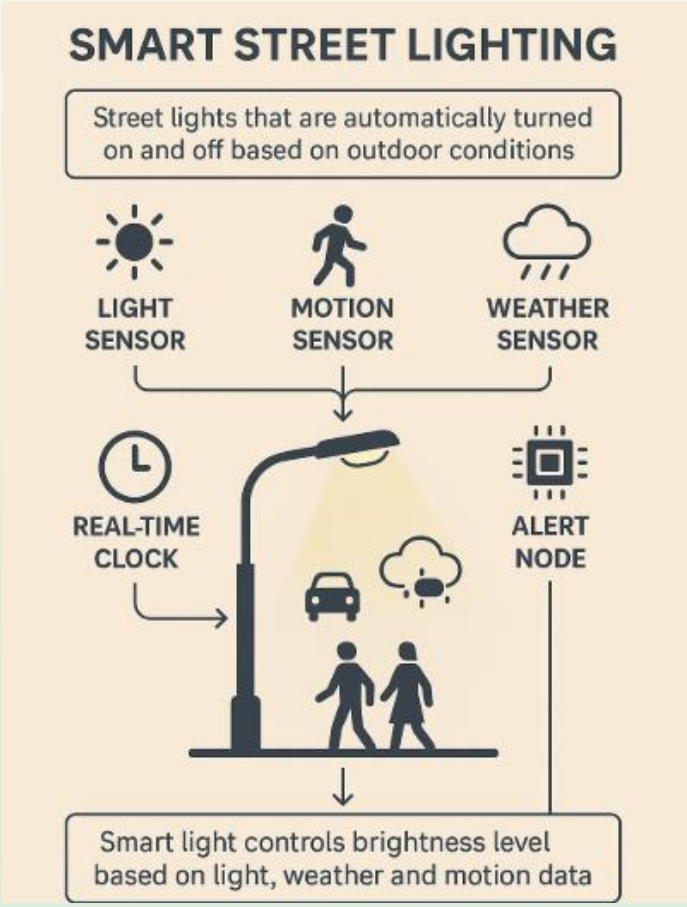


Parking Violation Detection

- Monitors no-parking zones using smart cameras and/or specialized readers.
- Detects illegal parking, captures evidence.
- Notifies the respect traffic authorities immediately.

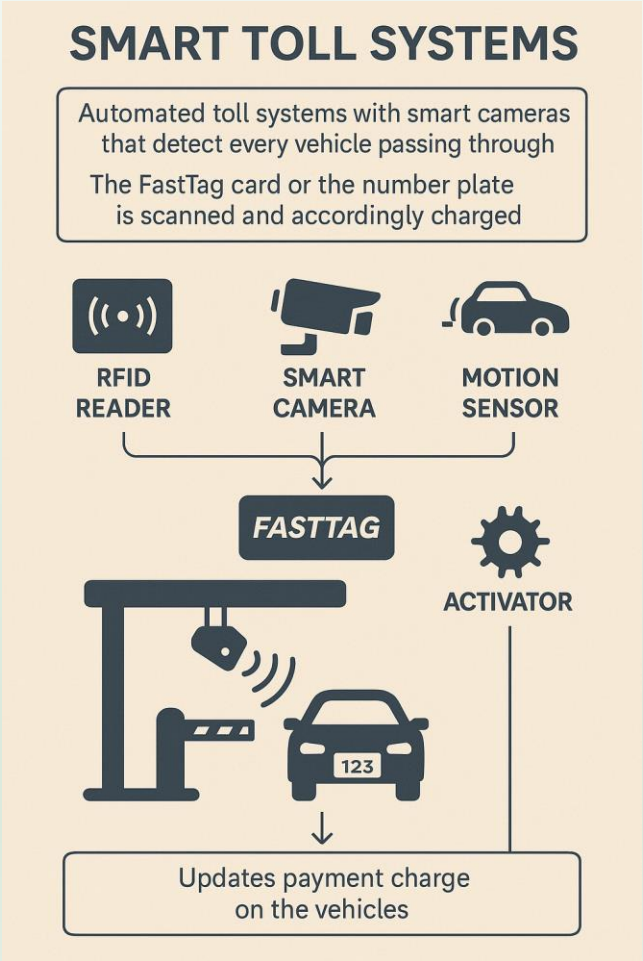


Smart Street Lighting



Device	Purpose
Light Sensor	To measure the intensity of light
Motion Sensor	To detect vehicles/pedestrian motion
Weather Sensor	To detect rain, fog, etc.
Real-Time Clock	Default settings of on and off
Smart Light	Controls brightness level
Alert Node	Sends lighting information for a specific region

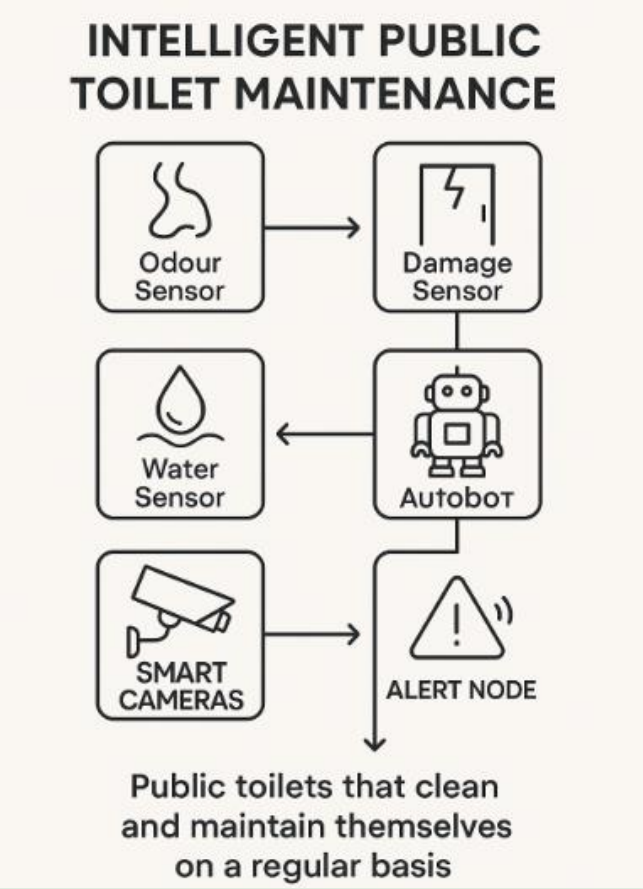
Smart Toll Systems



Device	Purpose
RFID Reader	To read FastTag card
Smart Camera	To read number plate through OCR
Motion Sensor	To specifically detect vehicles
Speed Sensor	To measure speed of vehicles
Activator	Activates cameras and readers upon arrival of vehicles
Alert Node	Updates payment charge on vehicles

Cleanliness

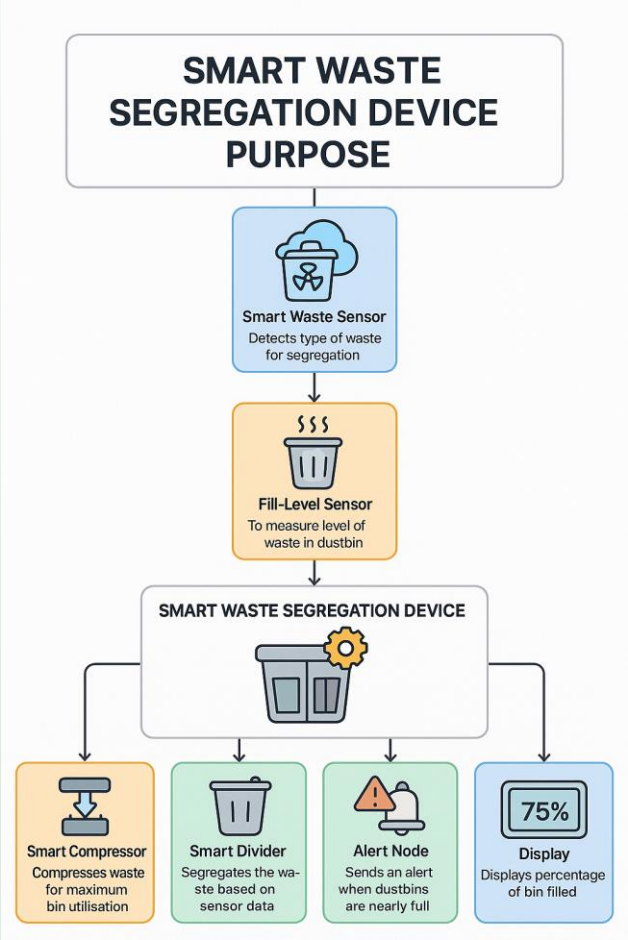
Intelligent Public Toilet Maintenance



Device	Purpose
Odour Sensor	To detect foul odour
Damage Sensor	To detect damaged doors/urinals/toilets/flush
Water Sensor	To detect water overflow
Dirt Sensor	To detect unnecessary dirt and untidiness
Smart Cameras	To recognise vandals (present outside)
Autobot	Makes necessary repairs
Alert Node	Alerts officials for repairs

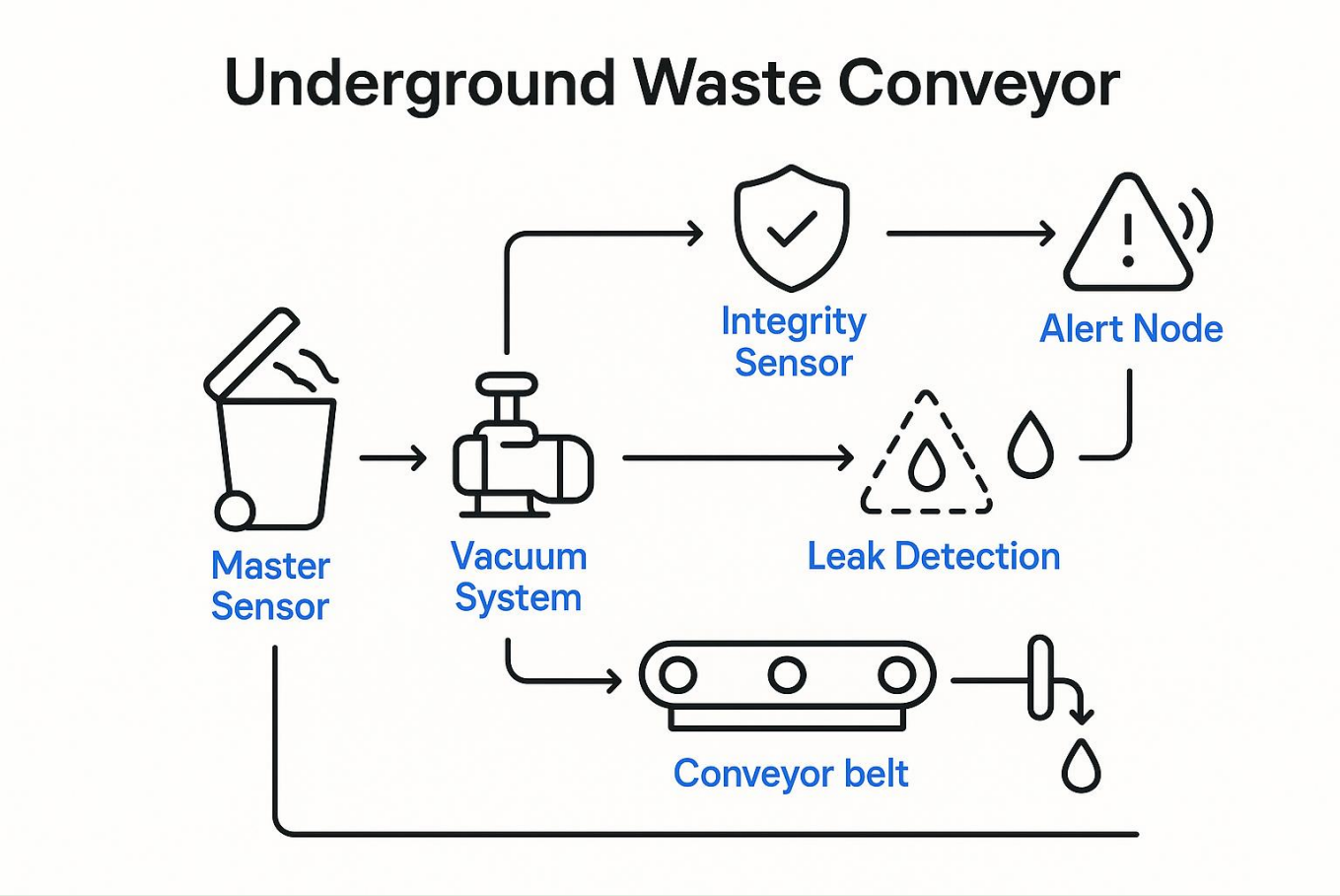
Smart Waste Segregation

- Throw waste into a singular public dustbin.
- It automatically segregates and separates the waste.
- When the dustbins reach a tipping point, officials are informed to collect and handle the waste.



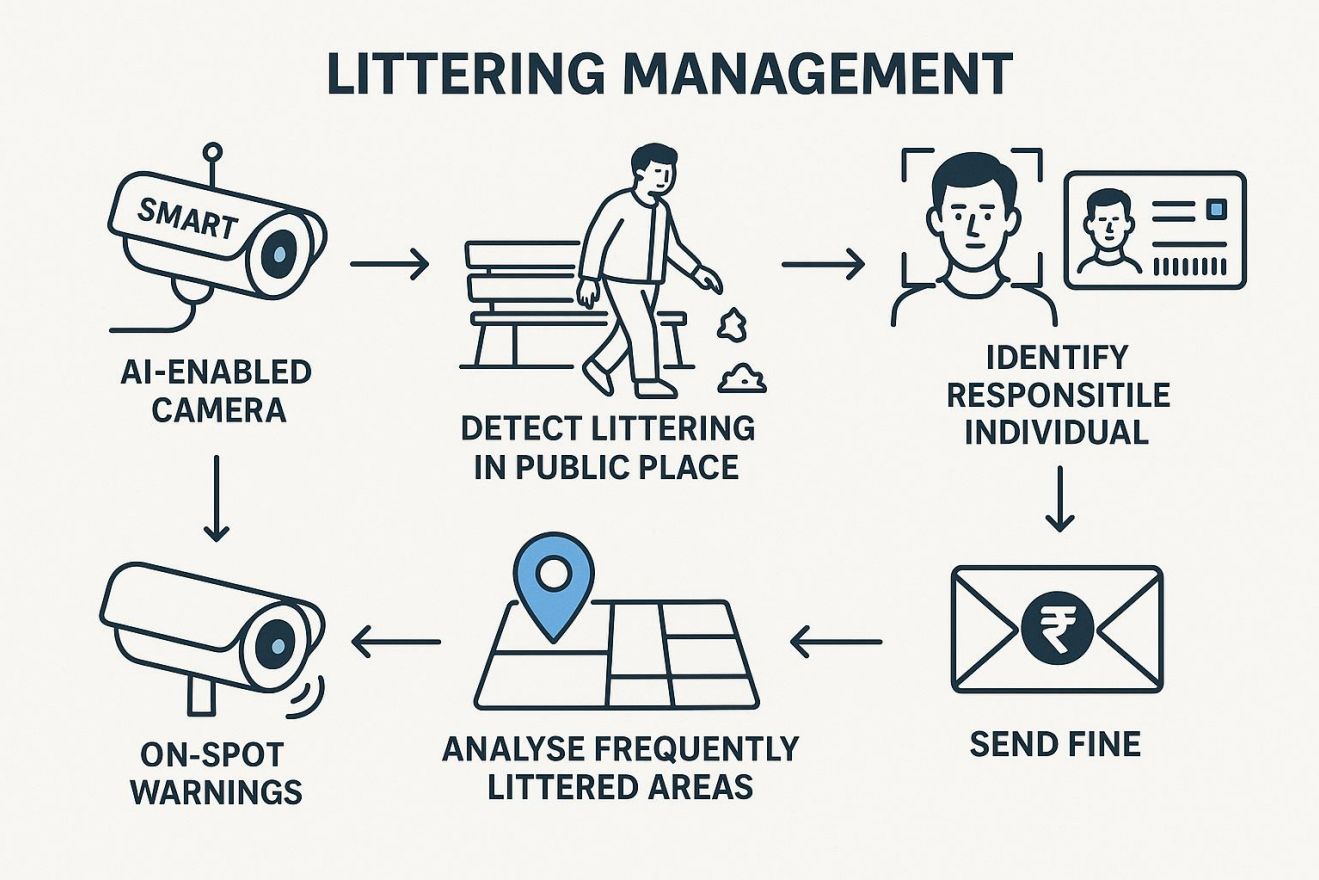
Underground Waste Conveyor

- Vacuum-powered underground pipeline system for waste transport.
- Sucks waste from nearby smart bins.
- Transfers it to a centralized unit, wherein recycling, composting, etc. take place.



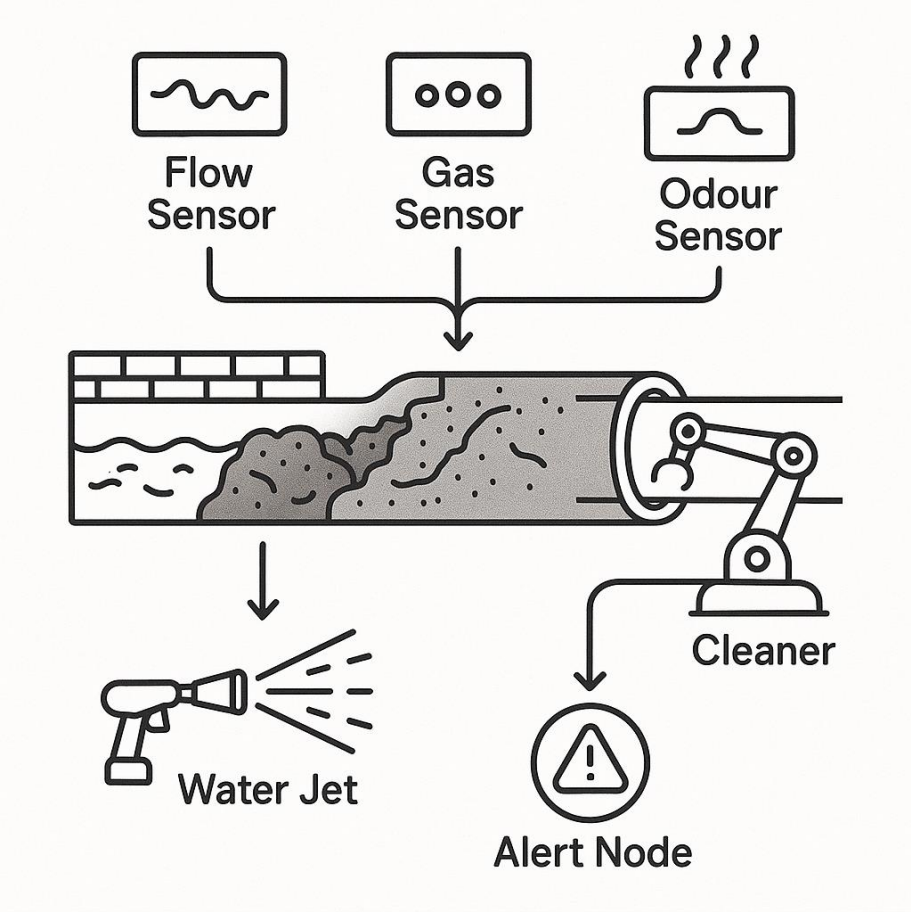
Littering Management

- Detect littering in public places.
- Identify the responsible individuals through their face and Aadhaar ID.
- Automatically send a fine.
- Analyse frequently littered areas.



Smart Drainage Cleaning

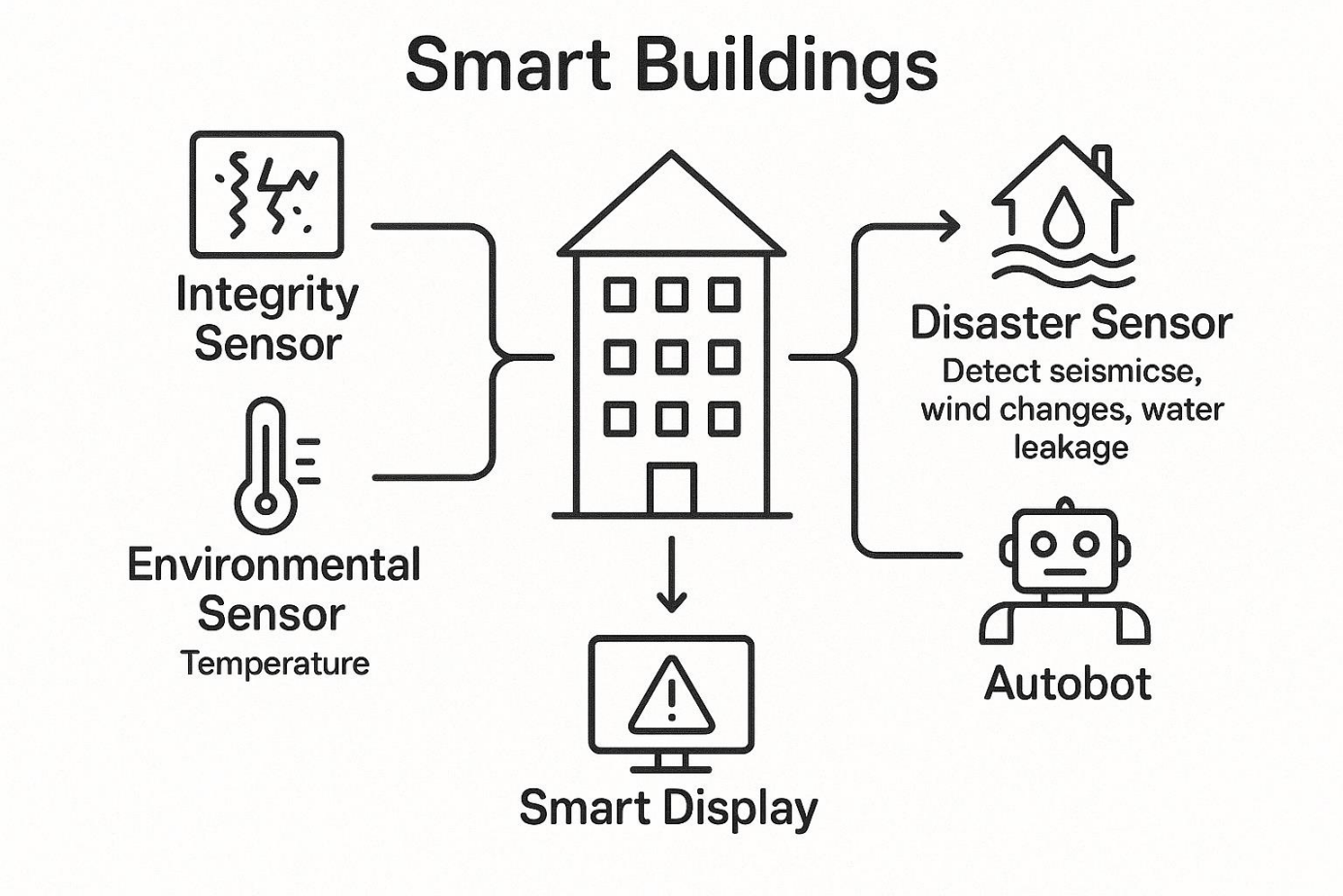
- Monitor drainage, detect blockages, corrosion, odour issues.
- Take appropriate action to fix the drainage.
- Notify required professionals if the same area is repeatedly affected.



Infrastructure

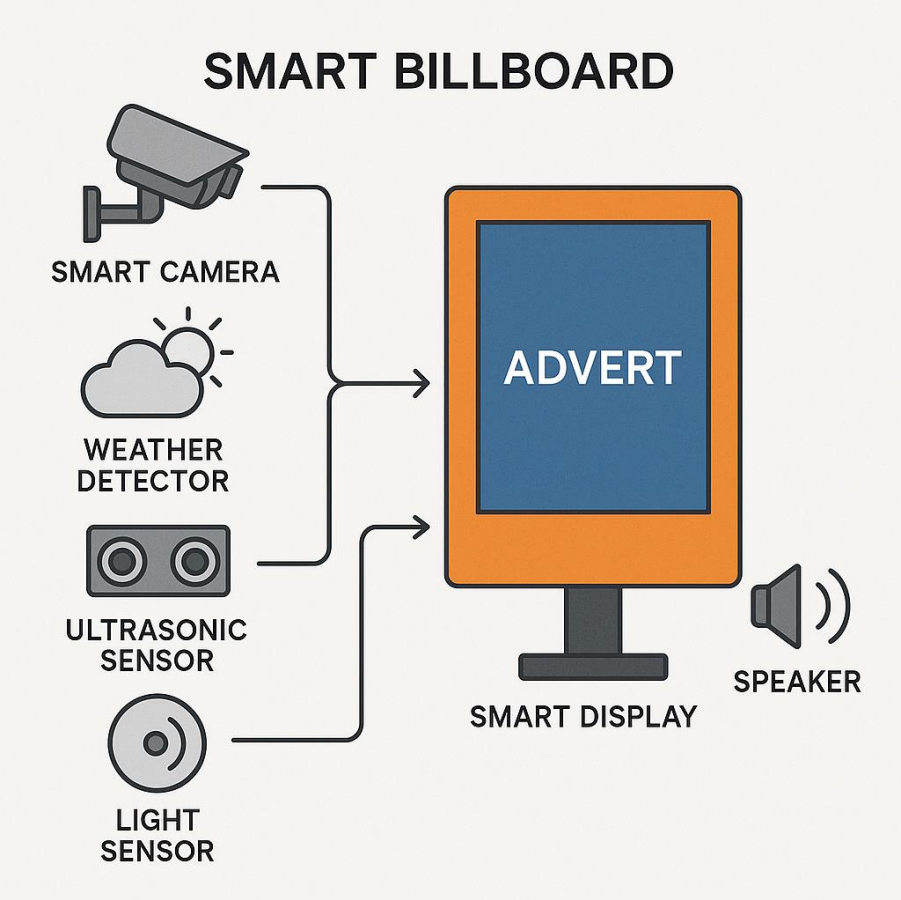
Smart Buildings

- Buildings that automatically measure their structural integrity.
- Automated repairs wherever possible.
- Notifications sent to officials for tasks that lie beyond automation.
- Also detect disasters such as earthquakes.



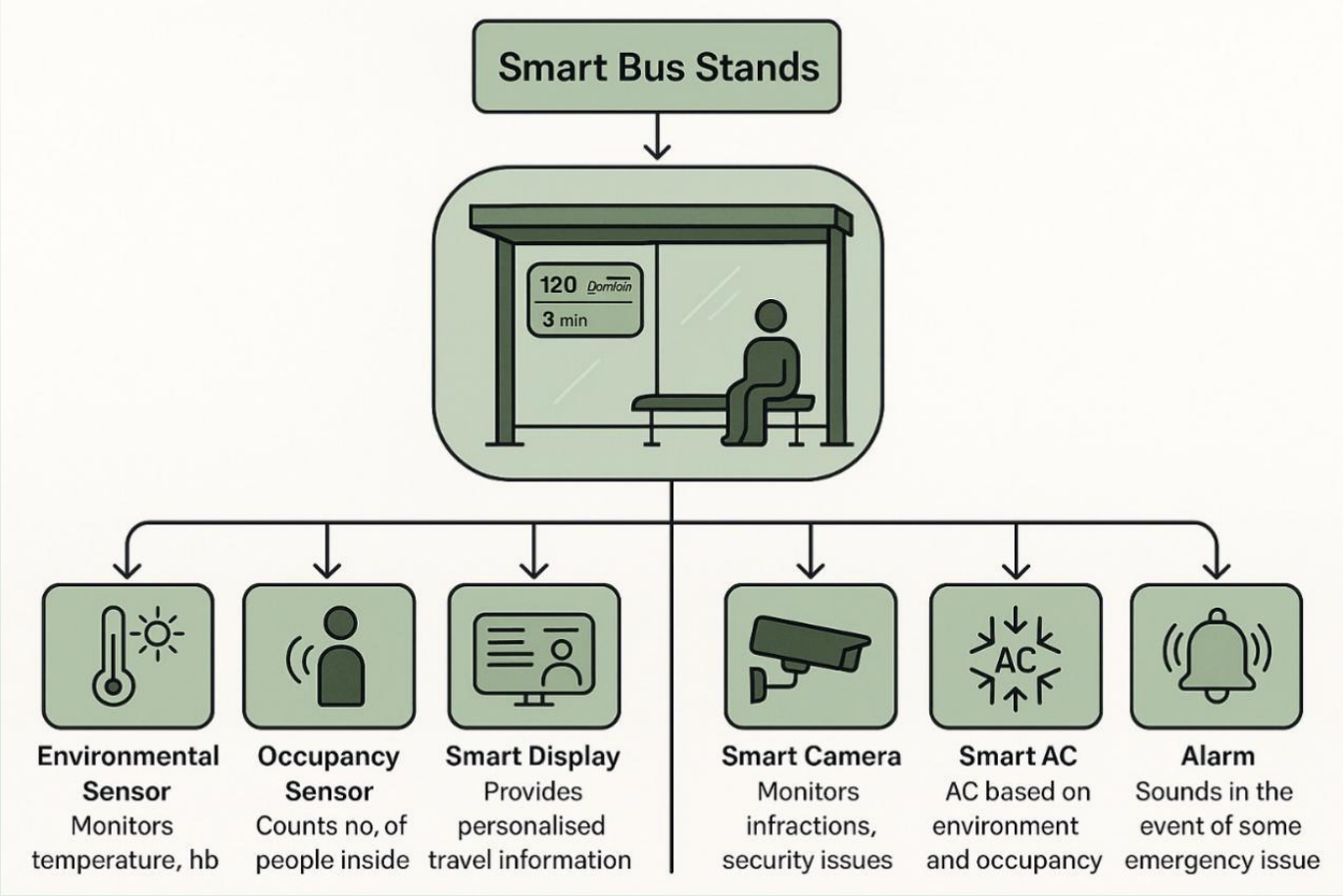
Smart Billboards

- Advertisement boards that don't need to be manually replaced every time.
- Based on the occasion and the environment, automatically provides a suitable advertisement.
- Updated remotely with new ads that the billboard itself decides when best to utilise.

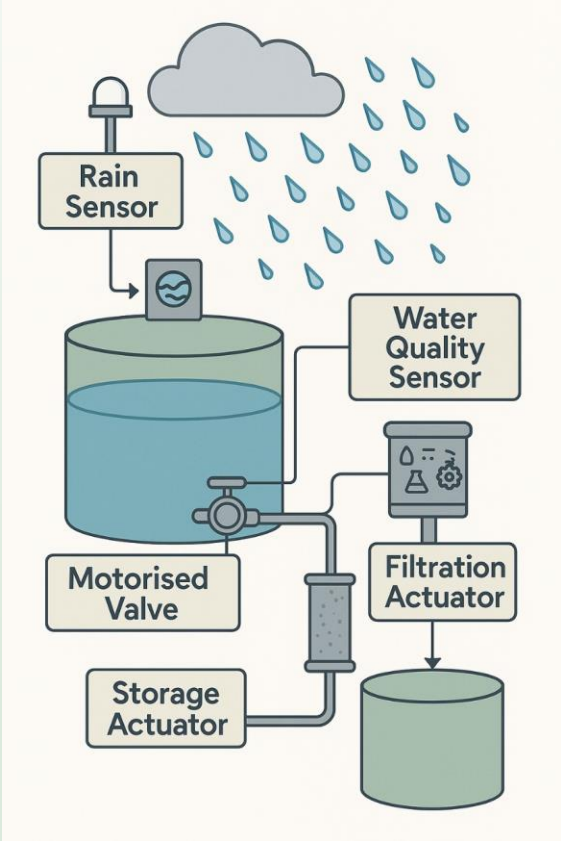


Smart Bus Stands

- Bus stands – small or large – as closed smart shelters.
- Provides travel information.
- Security and connectivity are ensured.
- Automatic temperature-based air conditioning.



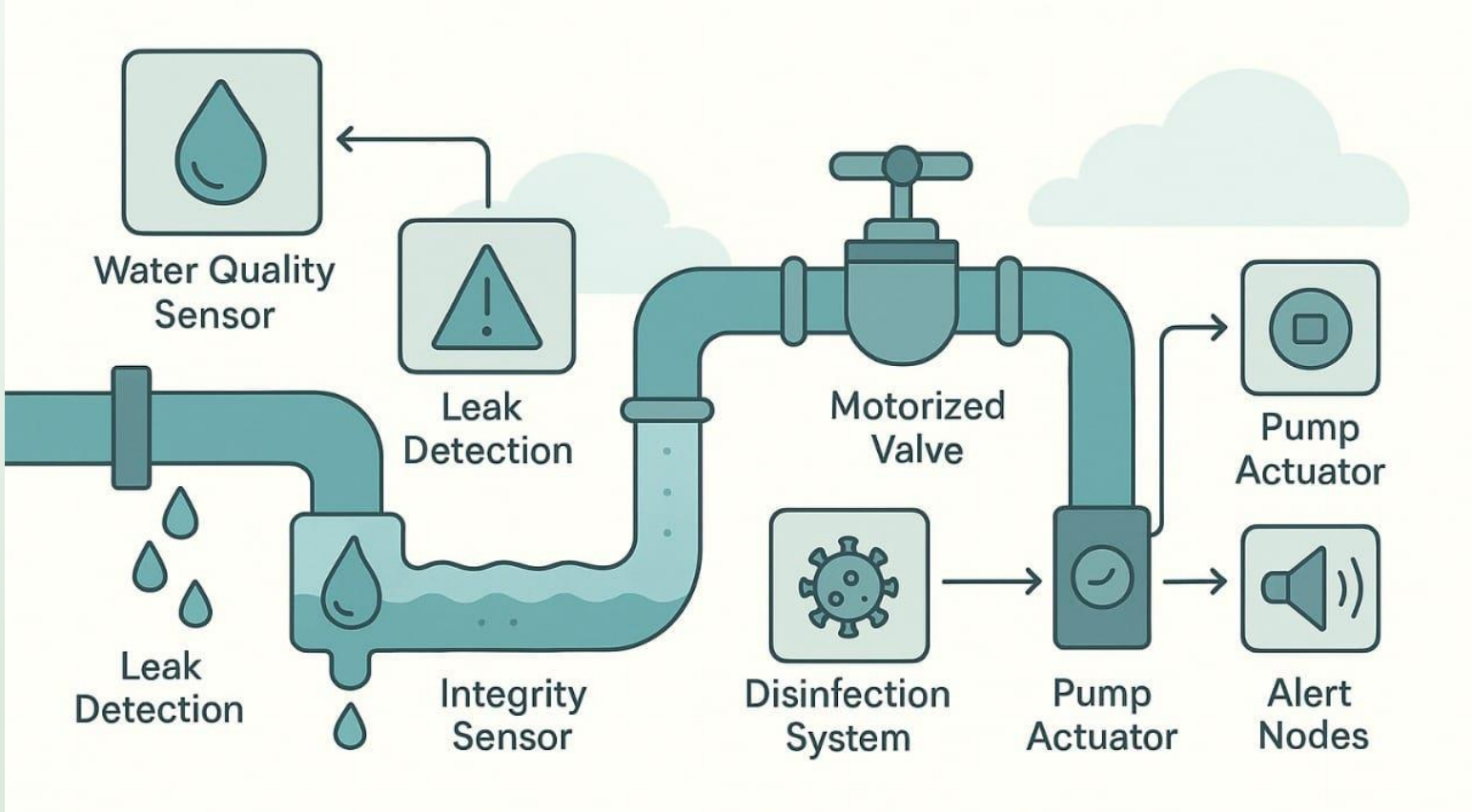
Rainwater Harvesting



Device	Purpose
Rain Sensor	To initiate harvesting process
Water Level Sensor	To monitor water level status in tank
Water Quality Sensor	To monitor pH, turbidity, etc.
Motorised Valve	Directs water flow to pipeline
Filtration Actuator	Performs necessary filtration
Storage Actuator	Redirects excess water to secondary storage

Water Quality Monitoring

- Monitors the quality of water delivered through metro pipelines.
- Detects leakage and implements precautionary measures.
- Sends real-time alerts to the required officials wherever necessary.



Thank You

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Yuvaraj K
Manobala P
Nithish S
Krishnaa S